

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Specification for Pig Traps
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Specification for Pig Traps

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R01	17.05.22	Issued for Review
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Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-011 PPL-SPC-011

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design

Table of Contents

1.	INTRODUCTION	5
1.1.	Background Information	5
1.2.	Purpose	6
1.3.	Abbreviations	7
2.	REGULATIONS, CODES AND STANDARDS	8
2.1.	Project Documents	8
2.2.	Shell Specifications	9
2.3.	International Codes and Standards	9
3.	ENVIRONMENTAL AND PIPELINE DESIGN DATA	10
3.1.	Environmental Data	10
3.2.	Pipeline Design Data	10
3.2.1.	24" Multiphase Pipeline Data	10
3.2.2.	36" Multiphase Pipeline Design Data	11
4.	RESPONSIBILITIES AND SCOPE OF SUPPLY	12
4.1.	Scope	12
5.	BASIC DESIGN, CONSTRUCTION AND RATINGS	13
5.1.	General	13
5.2.	PIG Trap System Component	13
5.3.	Supports	13
5.4.	Branch Connections	14
5.5.	End Closure	14
5.6.	Materials	15
5.7.	Fabrication	15
5.8.	Inspection and Testing	15
5.9.	Surface Preparation and Finish	15
5.9.1.	Surface Preparation	15
5.9.2.	Finish	16
5.10.	Pipeline Pig Passage Detector	16
5.11.	Nameplates and Labelling	16
6.	INFORMATION FOR MANUFACTURER/SUPPLIER	17

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

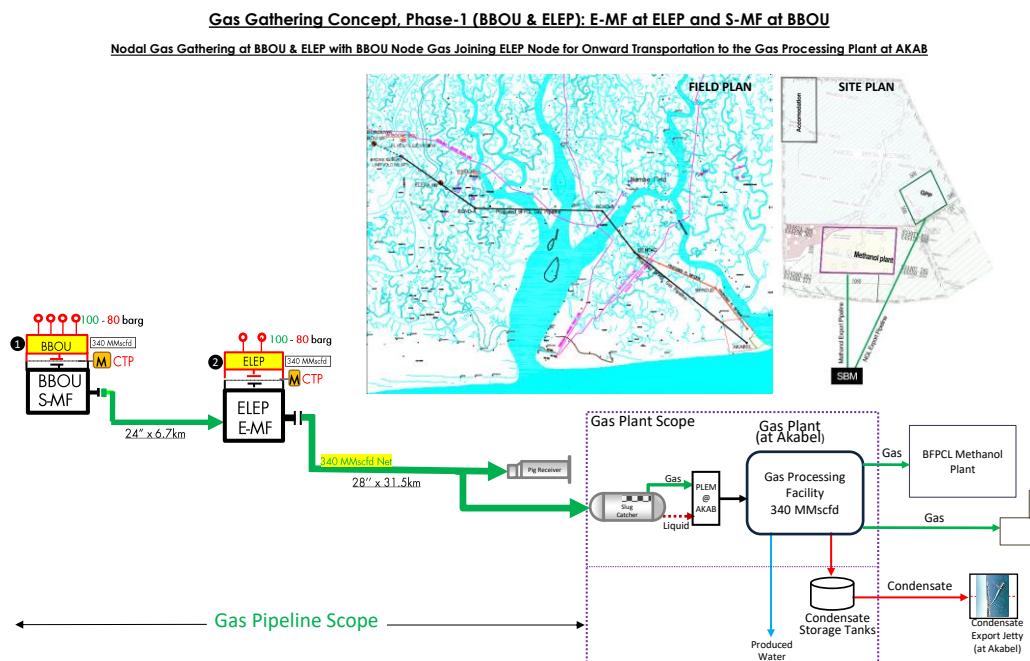


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This document covers the design, fabrication, inspection and testing requirement of pig traps to be used for the GAS PROCESSING AND METHANOL COMPLEX PROJECT. The pig traps as designed shall be considered to be part of the pipeline system to which it shall be connected.

This document specifies the requirements and gives recommendations for the design of pig launcher and receiver for this project. This document shall be read in conjunction with the document EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-005-Pipeline Equipment Data Sheets - Pig Launchers & Receivers.

The FEED shall fully comply with all specified relevant contractual requirements.

1.3. Abbreviations

Abbreviation	Meaning
End closure	Fitting, including a removable part or assembly, which provides quick and easy access to the major barrel when open and seals the bore when closed.
Kicker line (or bridle line)	Piping from the major barrel to the bypass line used to control the launch or receipt of a pig.
Main line	Major portion of a pipeline, between pig traps
Major barrel	Enlarged pipe section of a pig trap used for loading or retrieval of pigs.
Minor barrel	Pipe section of a pig trap between the reducer and the pig trap valve, of the same diameter as the pipeline
Pig	Device which can be propelled through a pipeline by fluid flow and is normally used for various internal activities such as separating fluids, cleaning and inspecting the pipeline. (A sphere is a spherically shaped pig).
Pig launcher	Pig trap for launching pigs.
Pig receiver	Pig trap for receiving pigs.
Pig signaller	Device set onto or into a pipe, which gives an indication of the passage of a pig.
Pig trap	Ancillary item of pipeline equipment, comprising a barrel, end closure and instruments, for introducing a pig into a pipeline or removing a pig from a pipeline. Pressure safety device (or tell-tale vent): safety device provided as part of the end closure door locking mechanism to safeguard personnel during door opening.
ANSI	American National Standard Institute
API	American Petroleum Institute
ASME	American Society of Mechanical engineers
AISC	American Institute of Steel Construction
AISI	American Institute of Iron and Steel
ASTM	American Society for Testing and Materials
OML	Oil Mining Lease
P&ID	Process and Instrument Diagram

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

Nigerian Local Standards and Codes

The Contract

Project Specifications

Employer's Standards

Shell DEPs

International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

2.1. Project Documents

S/N	Document number	Title
[1]	EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis for Design
[2]	EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005	Piping Class Specification
[3]	EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-015	Specification for Piping
[4]	EA-BFPCL-GPMC-GGPL-FEED-MPP-RPT-006	Pig Trap Calculation Note
[5]	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-001	Specification for Line Pipe
[6]	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-011	Specification for Pig Traps
[7]	EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Pipeline Wall Thickness Calculations
[8]	ANP-CW212280-ASNP2-LA-2105-00008	Data Sheet for Pig Traps
[9]	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003	Process Engineering Flow Sheets 1 to 8
[10]	EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-010	Datasheet for Pig Traps Enclosure
[11]	EA-BFPCL-GPMC-GGPL-FEED-GEN-PHY-001	Pipeline Design & Operating Philosophy
[12]	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-010	Pipeline Cleaning, Gauging, & Hydrotesting Specification.
[13]	EA-BFPCL-GPMC-GGPL-FEED-PPL-PHY-001	Pipeline Integrity Management Philosophy.

[14]	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-009.	Specification for Pipeline Construction.
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2.2. Shell Specifications

Ref	Document number	Title
[12]	DEP 31.38.01.24	Piping – Engineering and Layout Requirements
[13]	DEP 31.40.10.13-Gen (Feb. 2019)	Design of Pipeline Pig Trap Systems

2.3. International Codes and Standards

Ref	Document number	Title
[14]	API 5L	Specification for Line Pipe
[15]	ASME B31.8	Gas Transmissions and Distribution Piping Systems
[16]	ASME B16-47	Large Diameter Steel Flanges NPS 26 Through NPS 60
[17]	ASME Sec. VIII div.1 & DIV. 2	Boiler And Pressure Vessel code
[18]	ASME Sec. V	Non-Destructive Examination
[19]	ASME Sec. IX	Pressure Vessels Welding and Brazing Qualifications
[20]	ANSI B 16.5	Steel Pipe flanges and Flanged Valves and Fittings
[21]	ANSI B 16.9	Factory-Made Wrought Steel Butt-welding Fittings
[22]	ANSI B 16.21	Non-Metallic Flat Gaskets for pipe Flanges
[23]	ASTM A 694	Standards specifications for forging carbon and alloy steel, for pipe flanges, fittings, valves and parts for high pressure transmission service
[24]	ASTM A 350	Forged or Rolled Alloy-Steel Pipe Flanges, Forged Fittings and valves and Parts for High T. Service
[25]	ASTM A 516	Standard specification for pressure vessels plates, carbon steel, for moderate and lower temperature service
[26]	ASTM A 333	Standard specification for seamless and welded steel pipe for low temperature service
[27]	ASTM A 194	Carbon and Alloy Steel Nuts for Bolts for High Pressure and High T. Service
[28]	EN 10204	Material Certification
[29]	ISO 9001	Quality Management System

3. Environmental and Pipeline Design Data

3.1. Environmental Data

Refer for the basis of design document EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002

3.2. Pipeline Design Data

3.2.1. 24" Multiphase Pipeline Data

The 24" pipeline design data for the Multiphase Pipeline Project are as shown below

Parameter	Description
Design Code	ASME B31.8
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Design Life	30 years
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L-X70(PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength (SMYS)	483 Mpa
NPS	24"
Ultimate Tensile Strength	570 MPa
OD / Wall Thickness	15.88mm
Design Factor	0.6
Corrosion Allowance	3 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Cathodic Protection	Impressed current

3.2.2. 36" Multiphase Pipeline Design Data

The 36" pipeline design data for the Multiphase Pipeline Project are as shown below

PARAMETER	DESCRIPTION
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Design Life	30 years
Process Fluid	Condensate/hydrated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature	-13.3°C
Line Pipe Material Grade	API 5L-X70 (PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength	483 Mpa
NPS	36"
Ultimate Tensile Strength	570MPa
OD / Wall Thickness	23.83mm
Design Factor	0.6
Corrosion Allowance	6 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Cathodic Protection	Impressed current

4. Responsibilities and Scope Of Supply

4.1. Scope

Scope of work includes but not limited to design, materials, fabrication, inspection and testing, surface preparation and painting, documentation, preparation for shipment, preservation, and supply of Pig Trap Systems as elucidated below.

The Pig traps under the Gas gathering - Phase 1 project are follows;

Sn.	Size (in)		Pig Trap Type	Location	Pipeline System.
	Major Barrel	Minor Barrel			
1	28"	24"	Pig Launcher	BBOU manifold station	24" BBOU to Elepa Pipeline
2	28"	24"	Pig Receiver	Elepa Manifold Station	24" Pipeline from BBOU
3	40"	36"	Pig Launcher	Elepa Manifold Station	36" Elepa to Akabel Pipeline
4	40"	36"	Pig Receiver	Akabel Pigging Station	36" Pipeline from Elepa

5. Basic Design, Construction And Ratings

5.1. General

- The following description is intended to indicate the general and minimum requirement for the launcher and receiver and does not relieve the supplier of his full responsibility for fabrication performance and safety of the equipment.
- Trap assemblies shall be suitable for launching or receiving pigs for the purpose of inspection, gagging, cleaning and removing liquids from the pipeline which may contain water and/or liquid hydrocarbons and impurities such as sand and scale.
- Except lifting lugs, lifting facilities, such as runway beam, pig trays, may not be required unless demanded by the company.
- The corrosion allowance is as specified in the data sheets.
- -Internal diameter of the pig launcher and pig receiver assembly shall have a smooth continuous level surface.
- -Pig launcher shall have saddle mounting to create a fall towards, the pipeline to a minimum 1:100 unless otherwise specified on the PIDs.
- -Pig receiver shall have saddle mounting to create a fall towards, the trap door to a minimum 1:100. unless otherwise specified on the PIDs.

5.2. PIG Trap System Component

Barrel, reducer and spool

- The spool piece shall have the same nominal size as the connecting pipeline, flanged or bevelled end, as specified in requisition.
- Unless otherwise specified, the barrel shall have a sufficient length to accommodate three pipeline cleaning pigs.
- Barrels shall be designed in accordance with ANSI B 31.8
- A thermal relief line shall be provided at locations where shut-in pressure of trapped fluid could exceed the design pressure.
- The reducer should be eccentric for launcher and be concentric for receiver.

5.3. Supports

- Permanent supports shall be used to support and restrain the pig traps and these shall be designed to carry the weight of the pig trap system filled with water (or other fluids if their density is greater than that of water) together with the weight of pigs. The support under the barrel should normally be of the sliding/clamp type to compensate for expansion of the unrestrained part of the pipeline. The frictional force should be reduced to minimum by appropriate means.
- The height of the saddle should be such that the bottom of the barrel located 800 mm. above

ground level.

- Supports shall be positioned such that the pig trap valves can be removed for maintenance or replacement without removal of barrel.
- Design of supports shall comply with the requirements of ASME Sec.VIII
- Base plate, anchor bolts, etc. required for foundations shall be adequately design and all the applied forces and moments on the saddles shall be specified and provided for the manufacturer. Where necessary design data shall be requested from the manufacturer.

5.4. Branch Connections

- The trap shall be completed with all required connections.
- Weldolet and Thredolet are only allowed for connections equal or smaller than 2 inches. Connections larger than 2 inches shall be extruded outlets (reducing tees) or sweepolets if approved by company.

5.5. End Closure

- The end closure shall conform to the general requirements of ASME VIII division 1 section UG-35(b) (quick actuating closures). End closure door shall meet the requirements for a fail-safe design of the opening mechanism, specifically; the failure of any part of the opening mechanism shall leave the closure closed rather than open.
- The end closure shall be of the quick acting type, lever or hand wheel operated, and hinged vertically.
- The quick acting design should allow opening and closing by one man in approximately one minute, without the use of additional devices.
- The design of the end closure shall be suitable for permanent location in open environment.
- The activation of the seals shall be such that the fluid within the pig trap at any pressure between 1 bar (abs) and the pig trap design pressure be sufficient for this purpose.
- End closures with exposed screw expanders or captive ratchet braces should not be used, because of the high maintenance requirements and the non-fail-safe aspects of some opening mechanism designs.
- The end closure shall have the following safety devices:
 - a. A pressure locking device to prevent opening of the door when the pig trap is pressurized.
 - b. A safety bleeder that when released will alert the operator to a possible hazard unless pressure in the pig trap is relieved completely. Opening of the door shall not be possible unless the bleeder is released. Engaging the bleeder shall only be possible when the closure is closed. The bleeder shall be designed such that there is no risk of blockage. In very toxic service, the pressure safety device shall be non-venting.
 - c. The necessity for interlocking shall be decided by the Client.

5.6. Materials

Pipe for major and minor barrels shall be API 5L X70 unless otherwise specified in the specification for pipeline and/or the specification for pipes.

Flanges shall be weldneck type. All flanges and fittings shall be in accordance with the project requirement.

Studs and nuts shall be PTFE coated. The materials shall be ASTM A193/ grade B7 and ASTM A194/A194M grade 2HM

5.7. Fabrication

- Welding processes, welding procedures, welder qualifications, weld repairs, welding electrodes; thermal stress relief and heat treatment, etc. shall conform to the relevant project QAQC requirements.
- All main seam welds and nozzle to body welds shall be full penetration
- The inside of the trap shall be free from obstructions which could prevent the free rolling of spheres, or travel of pigs or carriers.
- Wall thickness transitions shall meet the welding configuration requirements as specified in the ANSI B31.8.

5.8. Inspection and Testing

Inspection and testing such as Visual Inspection, Dimensional check, 100% UT of all main weld etc. shall be performed before any coating or paint is applied with the approval of Owner.

All components shall be visually examined in accordance with ASME VIII division 1, part UG93.

Each weld on the trap and pig signaller shall be examined by 100% radiography (RT) and /or 100% magnetic particle examination (MT) or as may be required by the relevant project procedures.

For the end closure, all mating clamp and flange machined surfaces, door hinge, hinge attachments and locking mechanisms shall be subject to magnetic particle inspection. Any defects shall constitute a basis for rejection.

The trap shall be hydrostatically tested to 1.4 times the design pressure. The test pressure shall be held for a period of at least 4 hours. The acceptance criteria are no leakage or loss in pressure.

5.9. Surface Preparation and Finish

5.9.1. Surface Preparation

After acceptance of hydrostatic test, all external surfaces shall be prepared and prime coated in accordance with project blasting and painting procedures. The suppliers' proposed method of surface preparation shall be approved by the company.

5.9.2. Finish

All unpainted surfaces, e.g. flange surfaces, shall be properly protected against corrosion with anti-corrosion compound, easily removable by hydrocarbon solvents.

The pig launching and receiving trap including pig loading and unloading devices and also pig signaller shall be cleaned and shall be painted in accordance with project specification.

5.10. Pipeline Pig Passage Detector

The signaller shall be intrusive type and mounted on the trap/pipeline via a DN 50 branch connection.

For pipelines which cannot be depressurized for pig signaller maintenance, the signallers shall be complete with a ball valve to isolate the pig signaller from the pipeline and a portable jacking tool for safe lifting of the transfer mechanism.

5.11. Nameplates and Labelling

Each pig launching and receiving trap shall be labelled with engraved stainless or non-corrosive alloy nameplates together with non-corrosive fixing materials, showing all data as called for in this specification including but not limited to, the followings:

- d. Company name and order number.
- e. The year of manufacture.
- f. Manufacturer's name or trade mark.
- g. Type of materials and size
- h. Flange pressure rating.
- i. Design and test pressure.
- j. Dimensions and physical properties including weight.
- k. Tag number.
- l. Design temperature.

6. Information For Manufacturer/Supplier

Further to the information included in other parts of this specification, equipment data sheet shall be completed and furnished with the requisition as a part of Appendix of this specification document.